

ASSESSMENT TOOL 1 - Short Answer Test

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16. A high-resolution image consumes excessive storage and processing time. Write a Python program to reduce image size.

AIM:

To reduce the size (resolution) of a high-resolution image using OpenCV, thereby decreasing storage space and processing time.

ALGORITHM:

Import the required libraries (`cv2`).
Read the input image.
Display the original image dimensions.
Calculate the new width and height (e.g., 50% of the original size).
Resize the image using `cv2.resize()`.
Save the resized image.
Display both the original and resized images.

CODE:

```
import cv2
img = cv2.imread("input.jpg")
print("Original Shape:", img.shape)
new_width = int(img.shape[1] * 0.5)
new_height = int(img.shape[0] * 0.5)
resized = cv2.resize(img, (new_width, new_height))
print("Resized Shape:", resized.shape)
cv2.imwrite("resized_image.jpg", resized)
cv2.imshow("Original Image", img)
cv2.imshow("Resized Image", resized)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT:

```
TEST 1.py - C:/Users/Kaviya V/Downloads/TEST 1.py (3.14.3)
File Edit Format Run Options Window Help

import cv2

# Read image
img = cv2.imread("input.jpg")

# Display original size
print("Original Shape:", img.shape)

# Resize image to 50%
new_width = int(img.shape[1] * 0.5)
new_height = int(img.shape[0] * 0.5)

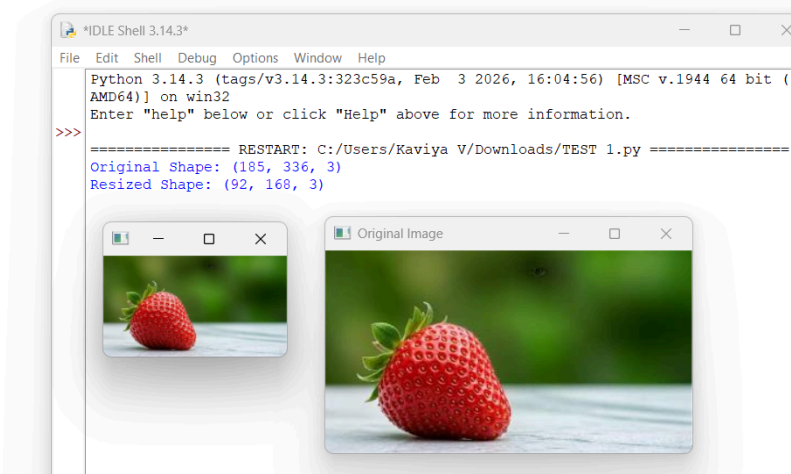
resized = cv2.resize(img, (new_width, new_height))

print("Resized Shape:", resized.shape)

# Save resized image
cv2.imwrite("resized_image.jpg", resized)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Resized Image", resized)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/Kaviya V/Downloads/TEST 1.py =====
Original Shape: (185, 336, 3)
Resized Shape: (92, 168, 3)
```

Result:

Thus, the high-resolution image was successfully resized using OpenCV.

17. An image captured under uneven lighting shows poor contrast. Write a Python program to improve image contrast.

AIM:

To improve the contrast of an image captured under uneven lighting using Histogram Equalization.

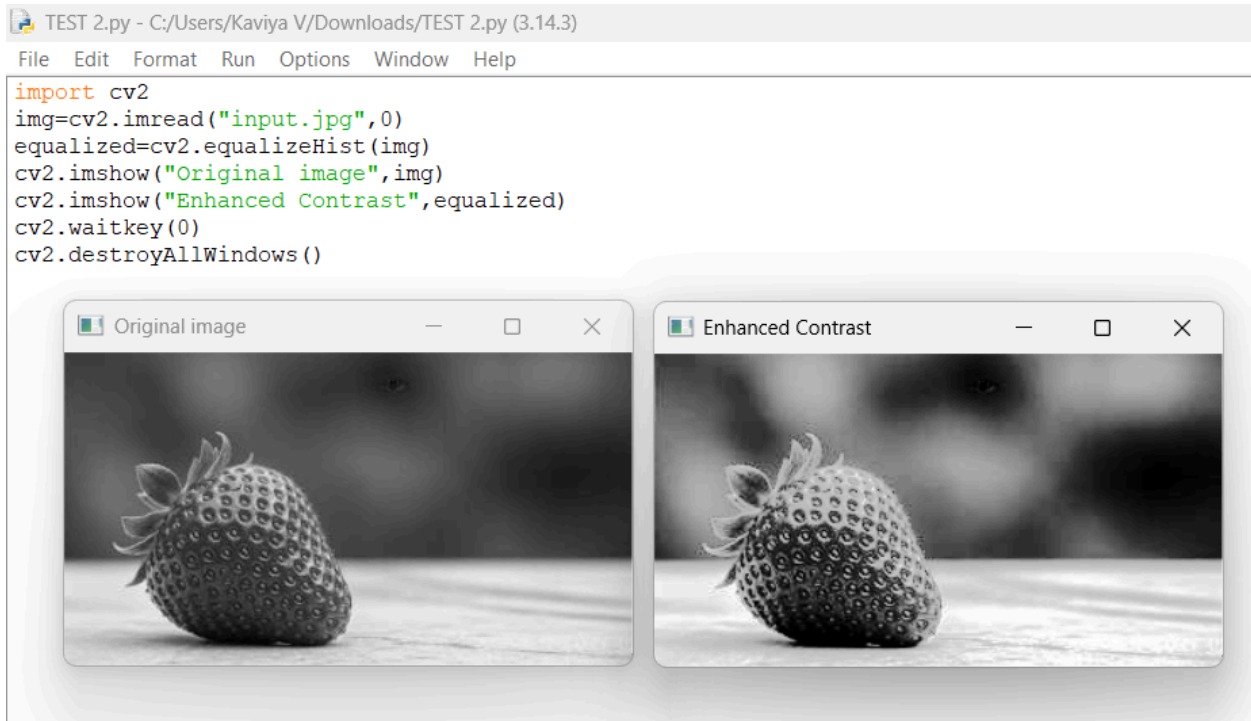
ALGORITHM:

- Import the required libraries.
- Read the input image in grayscale.
- Apply histogram equalization using `cv2.equalizeHist()`.
- Display both the original and enhanced images.
- Compare the improvement in image contrast.

CODE:

```
import cv2
img=cv2.imread("input.jpg",0)
equalized=cv2.equalizeHist(img)
cv2.imshow("Original image",img)
cv2.imshow("Enhanced Contrast",equalized)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT:



Result:

Thus, the image contrast was successfully improved using Histogram Equalization.

18. A medical image has improper quantization. Write a Python program to demonstrate quantization effects.

AIM:

To demonstrate the effect of image quantization by reducing the number of gray intensity levels in a grayscale image.

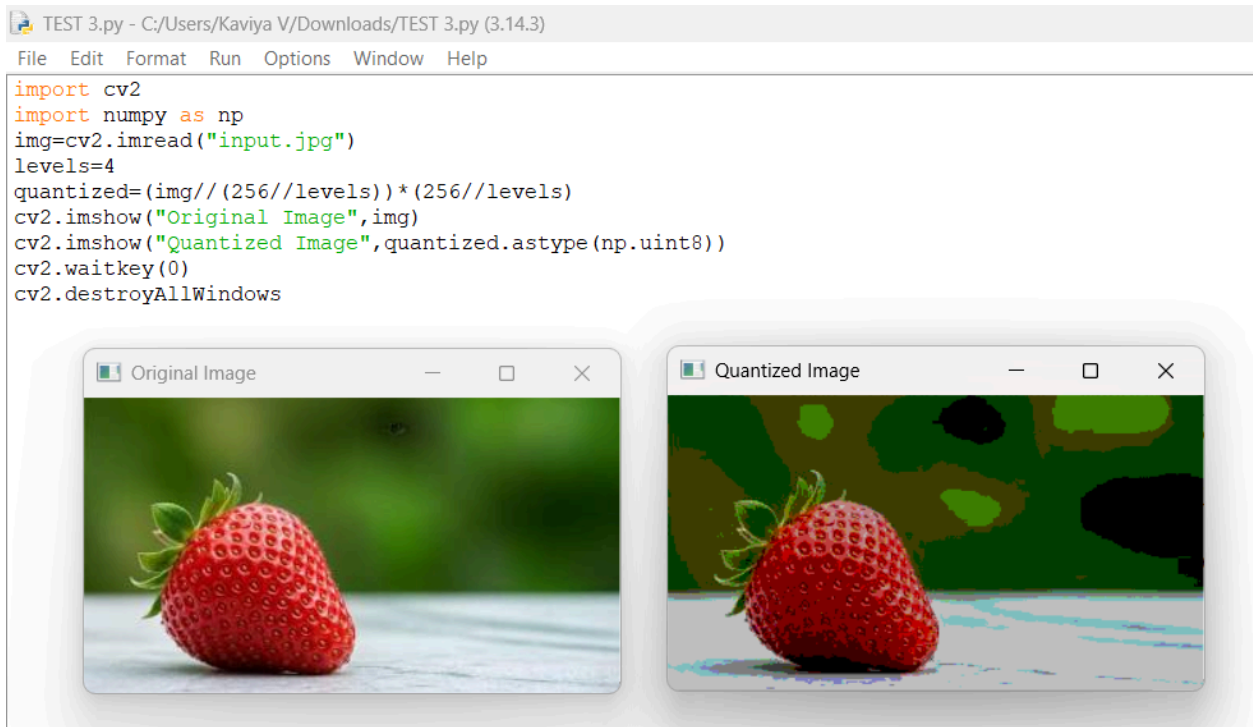
ALGORITHM:

- Import the required libraries (`cv2` and `numpy`).
- Read the grayscale image.
- Select the desired number of quantization levels.
- Reduce the intensity levels using the quantization formula.
- Convert the quantized image to unsigned integer format.
- Display both the original and quantized images.

CODE:

```
import cv2
import numpy as np
img=cv2.imread("input.jpg")
levels=4
quantized=(img//(256//levels))*(256//levels)
cv2.imshow("Original Image",img)
cv2.imshow("Quantized Image",quantized.astype(np.uint8))
cv2.waitKey(0)
cv2.destroyAllWindows
```

OUTPUT:



Result:

Thus, image quantization was successfully demonstrated by reducing the number of gray intensity levels.